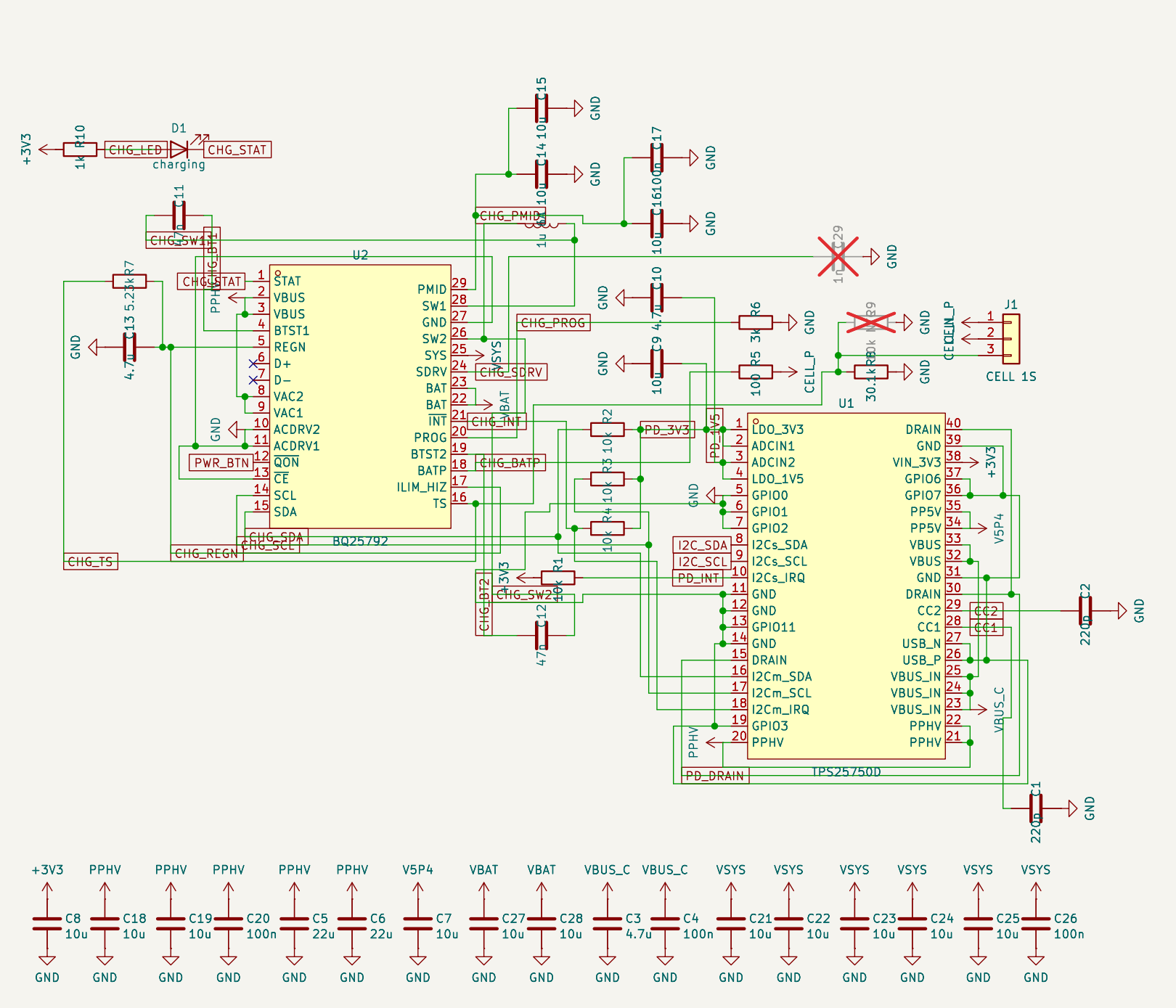
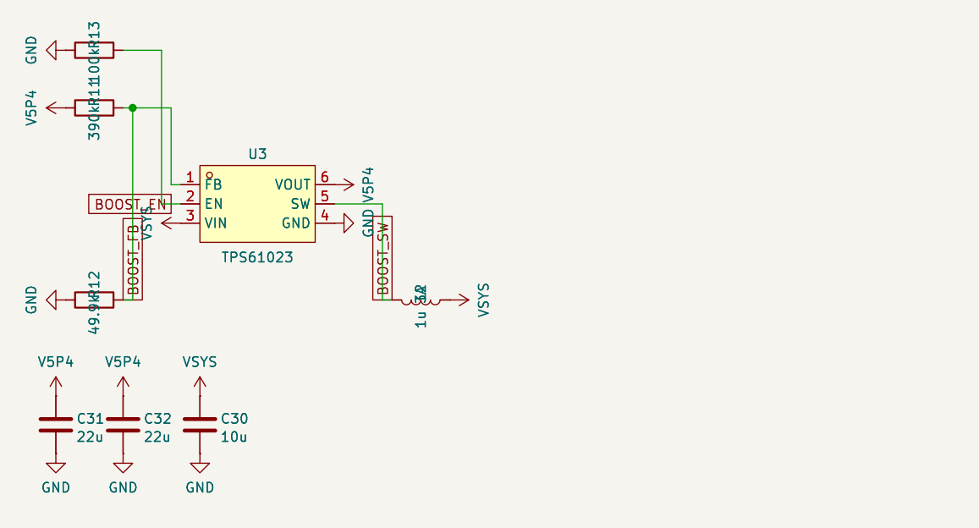
Battery, charger and phone power

One USB-C port does everything: TPS25750D (USB PD) with BQ25792 behind its bidirectional PPHV path, as TI's Figure 10-2. Charger plugged in: sink, BQ25792 charges the cell. Phone plugged in: the RT1011 sees a USB host, tells the TPS25750 to stop sinking, and on request swaps power role so the BQ25792 (OTG, 5 V up to 3 A) charges the phone. Data stays phone = host. ADCIN1 = LDO_3V3, ADCIN2 = LDO_1V5: AlwaysEnableSink (a flat pod still charges), I2C address #1. Patch bundle loaded by the RT1011 over I2C at boot; the TPS25750 drives the BQ25792 on its I2Cm bus. The cell reaches BAT through a ship FET (see Power button and ship FET).



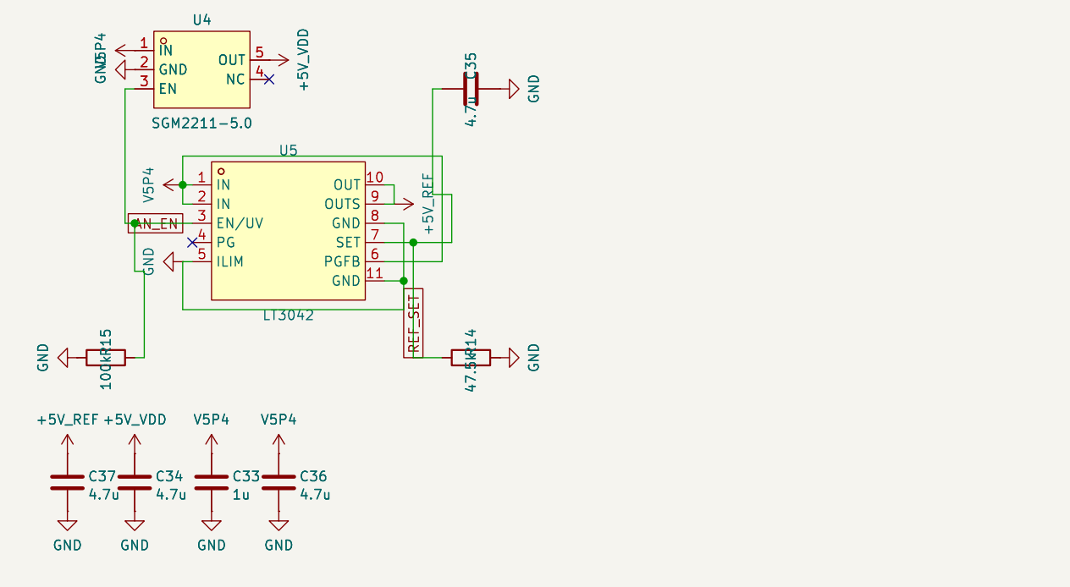
5.25 V boost

TPS61023: VOUT = 0.595 V x (1 + 390k/49.9k) = 5.25 V, under the RT1011's 5.5 V USB_OTG1_VBUS limit with tolerances, 0.5 V over the LT3042. Feeds the DAC's 5 V LDOs and the amp rails.



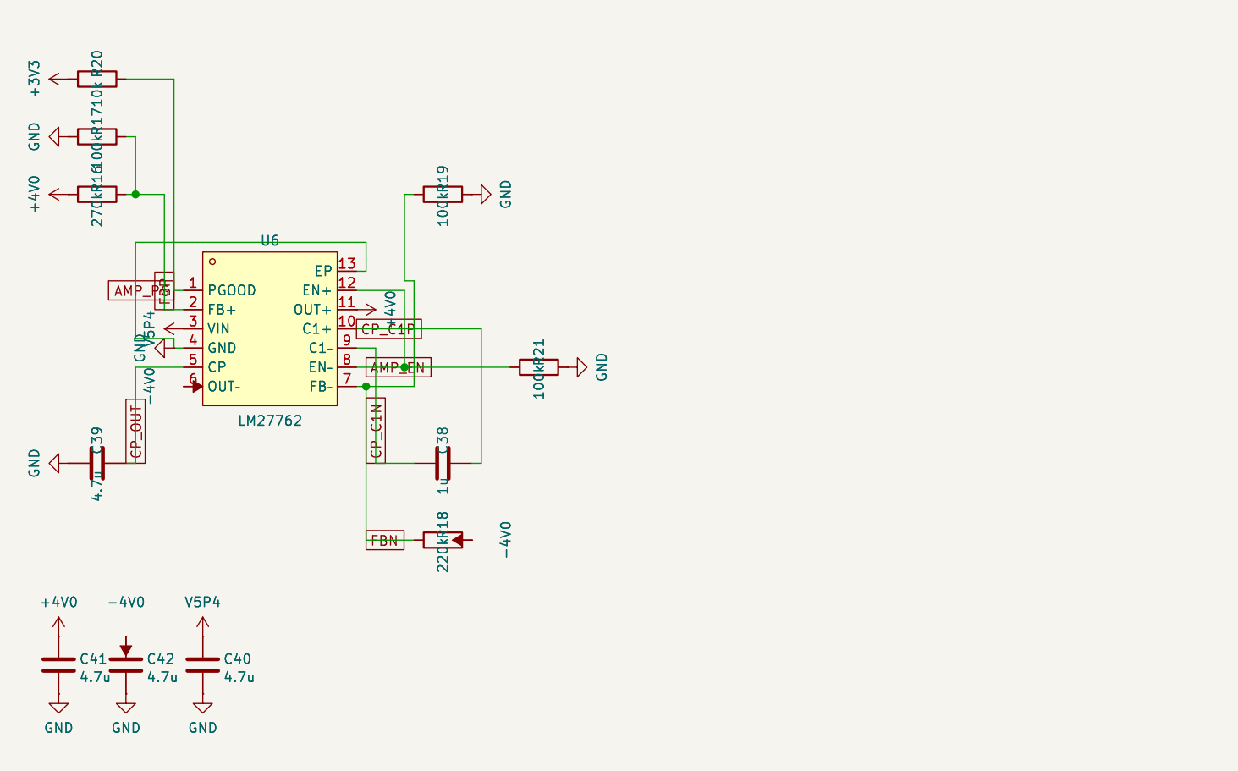
DAC 5 V supply and reference

SGM2211-5.0: VDDL/VDDR. LT3042: VREFH, 100 uA x 47.5k = 4.75 V, below VDDL/R with tolerances (AK4493S: VREFH from VDDL/R – 0.5 V to VDDL/R). Thin film, 0.5 %, 50 ppm.



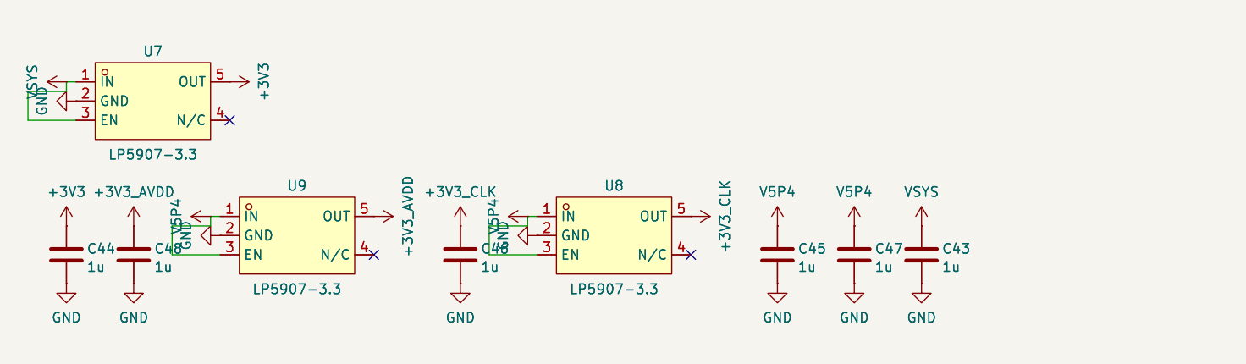
Amp rails +/-4 V

LM27762: +1.2 V x (270k+100k)/100k = +4.44 V, -1.22 V x (220k+100k)/100k = -3.90 V. TI: R2 and R4 (FB to GND) no less than 50k (datasheet 7.2.2.1, 7.2.2.3). The OPA1612's inputs work up to V+ – 2 V and its + input peaks near 1.9 V at 0 dBFS, so +4.44 V (the positive LDO runs from V5P4 with 45 mV dropout). The negative rail stays under 4.5 V for the charge pump's headroom.



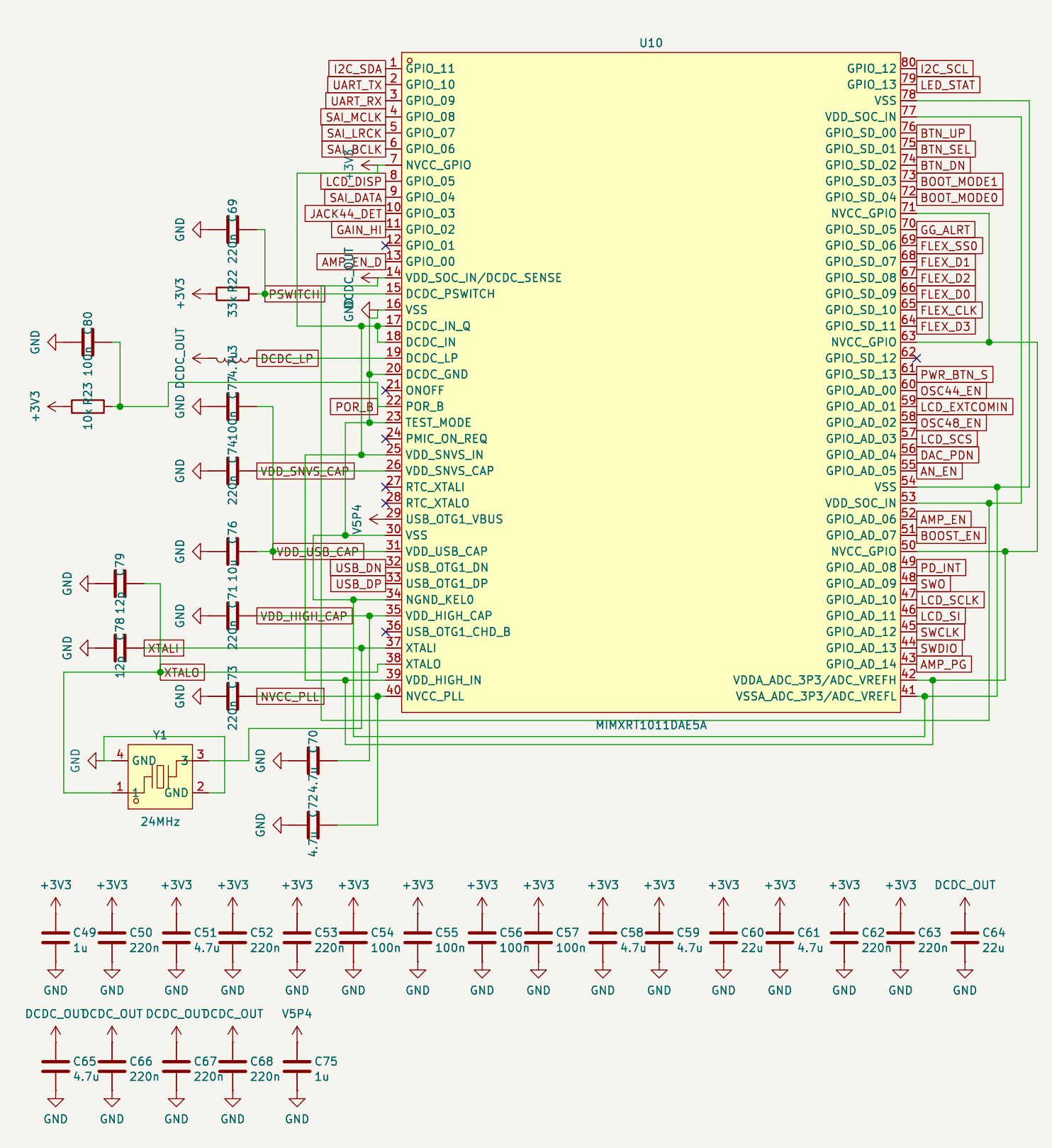
3.3 V rails

Three LP5907s: digital from VSYS (always on); clocks and DAC AVDD from V5P4, so they stay in regulation when the cell sags below 3.4 V, and are off with the rest of the audio side.



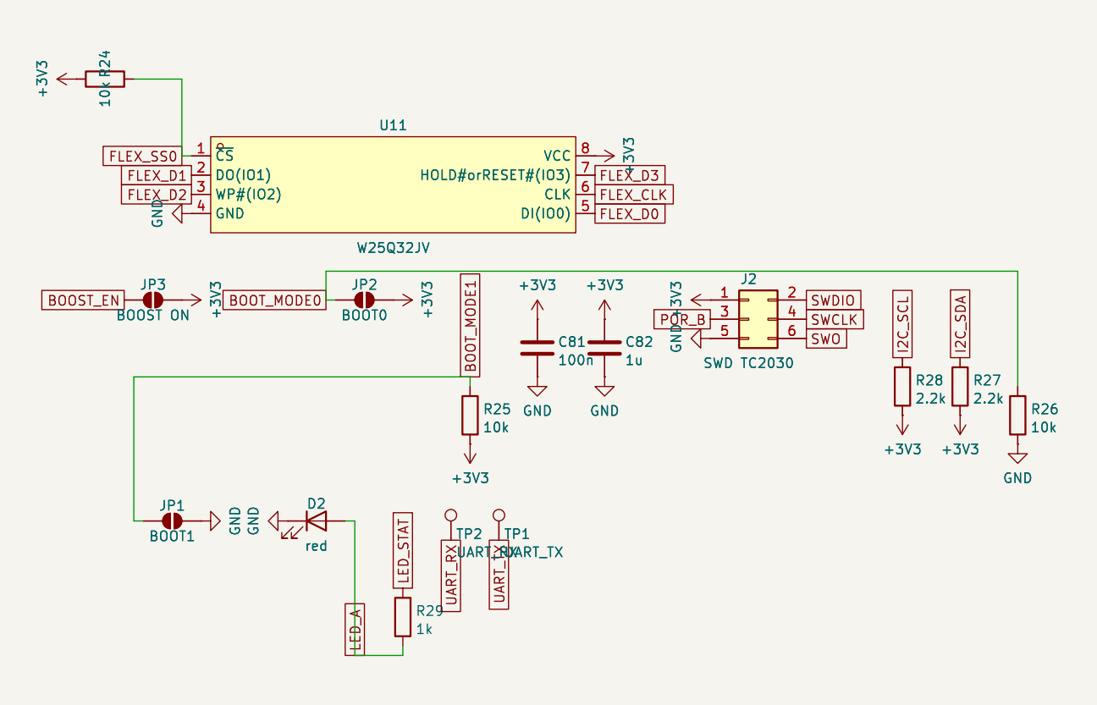
RT1011 MCU

Power, crystal, reset and DCDC follow the open Makerdiary iMX RT1011 Nano Kit (CC BY-SA). No 32.768 kHz crystal: the RTC runs from the internal oscillator.



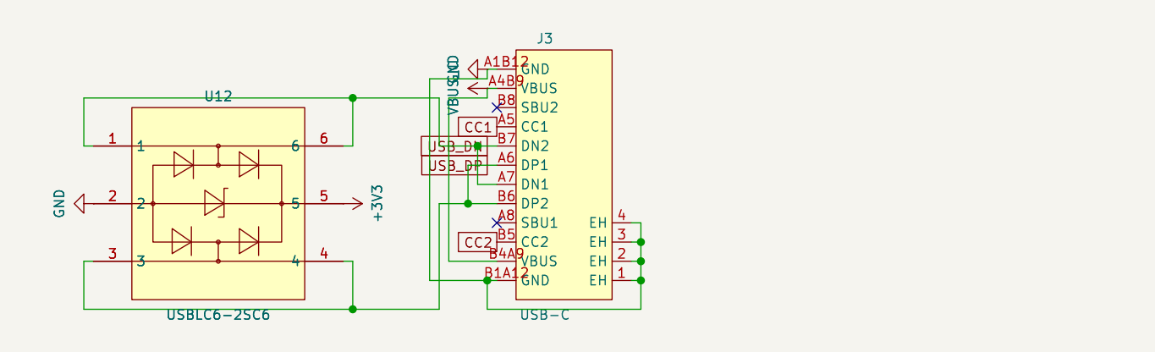
Boot flash, boot mode, debug

Bridge JP1 and JP2 for the serial downloader; open: internal boot from flash. The USB PHY runs from V5P4, and a blank RT1011 leaves the boost off: bridge JP3 too for the first USB flash, or flash over SWD. SWD and reset on a Tag-Connect TC2030 footprint; UART on test pads.



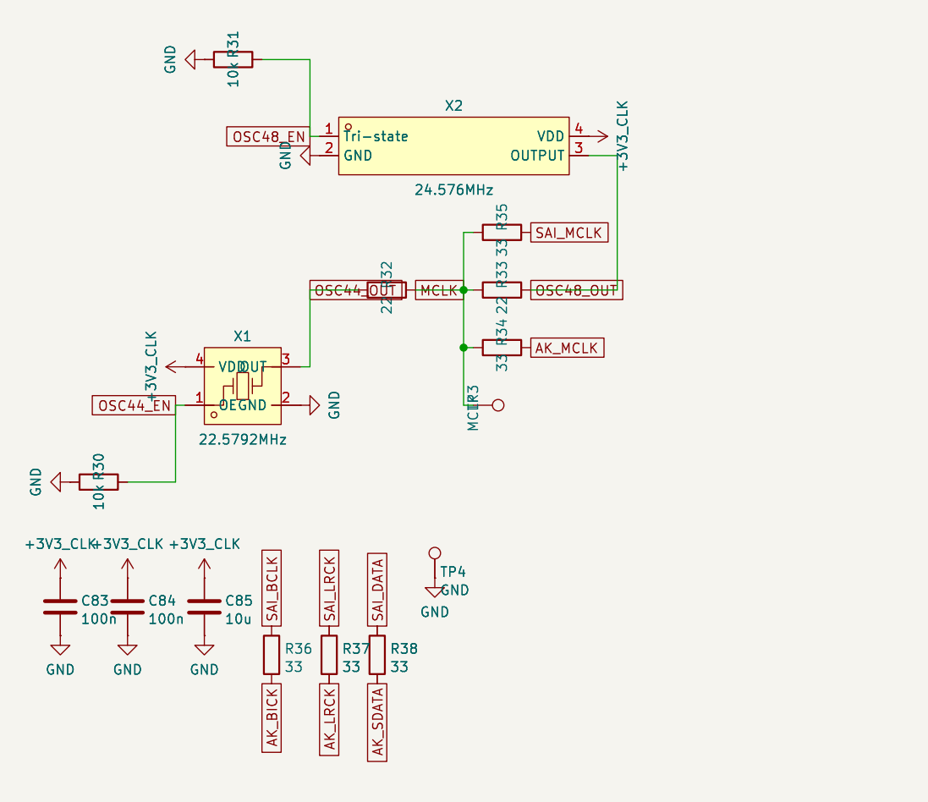
Phone USB-C

The pod's one port: a short C-to-C cable to the phone, or a charger. USB 2.0 data to the RT1011. The ESD clamp's reference is +3V3, not VBUS, which can be up to 9 V when charging.



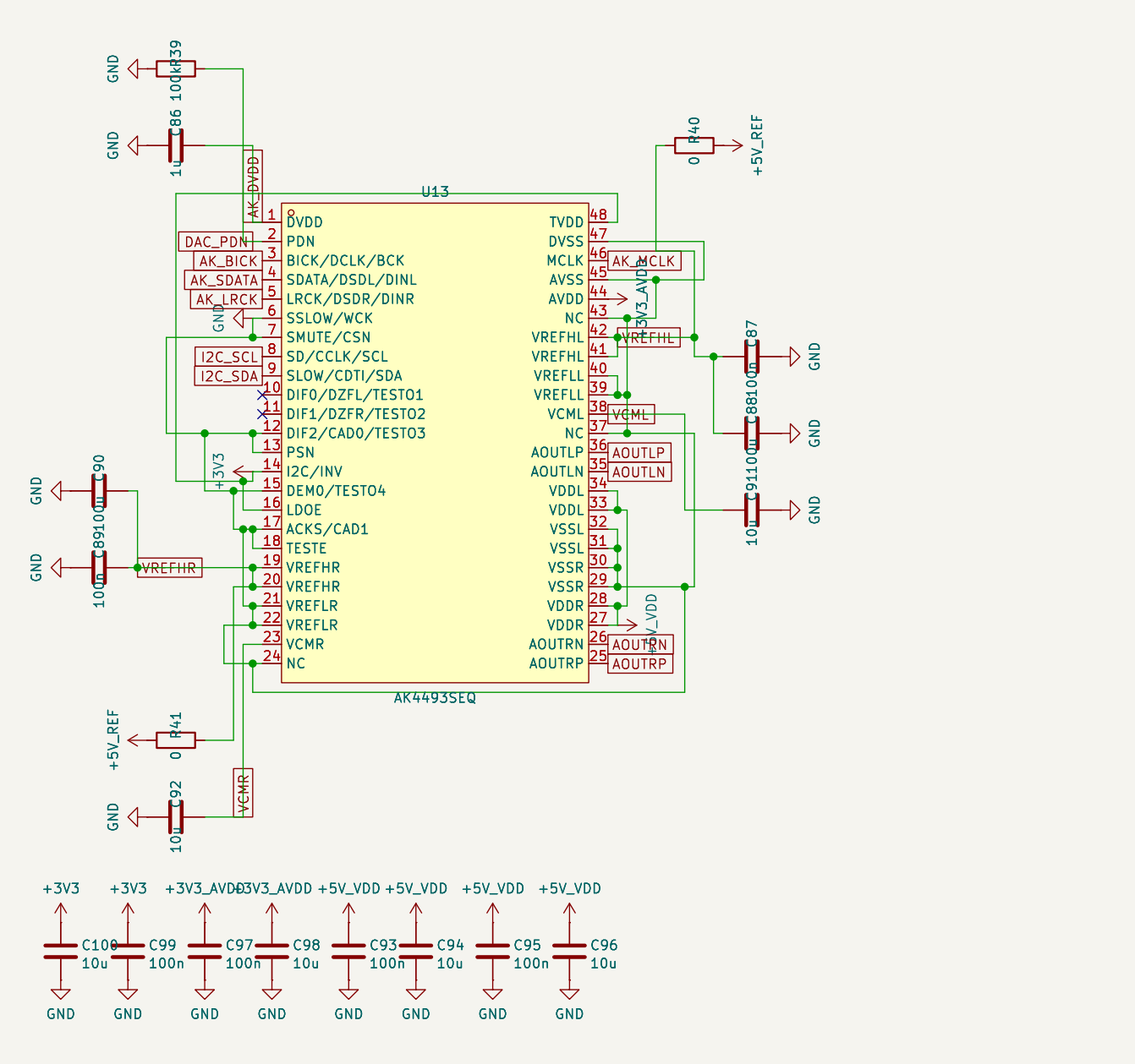
Audio clocks

One oscillator runs at a time (OE high). 22 ohm at each output, then damping resistors to the DAC and the SAI's MCLK input. SAI is clock slave to MCLK.



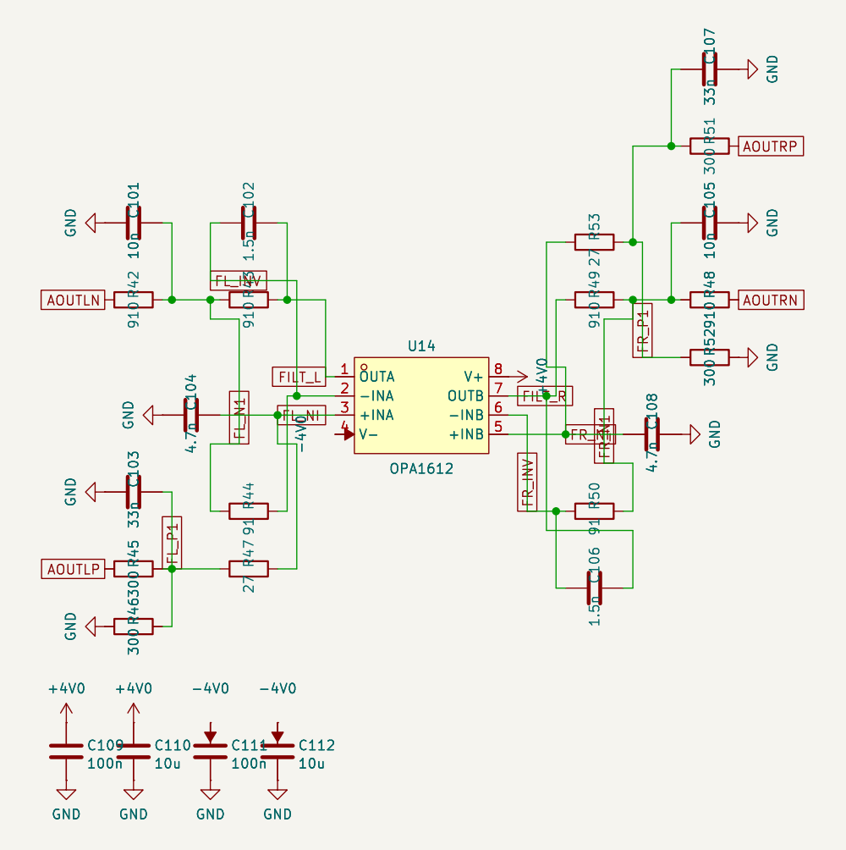
AK4493SEQ DAC

Register control over I2C (address 0x10), internal LDO on. Follows AKM Figure 78, with 100 uF ceramic in place of 470 uF on each VREFH. VREFH straight from the LT3042 (0 ohm links; AKM's 10 ohm is for a noisy supply, p. 98): no RC corner on the reference.



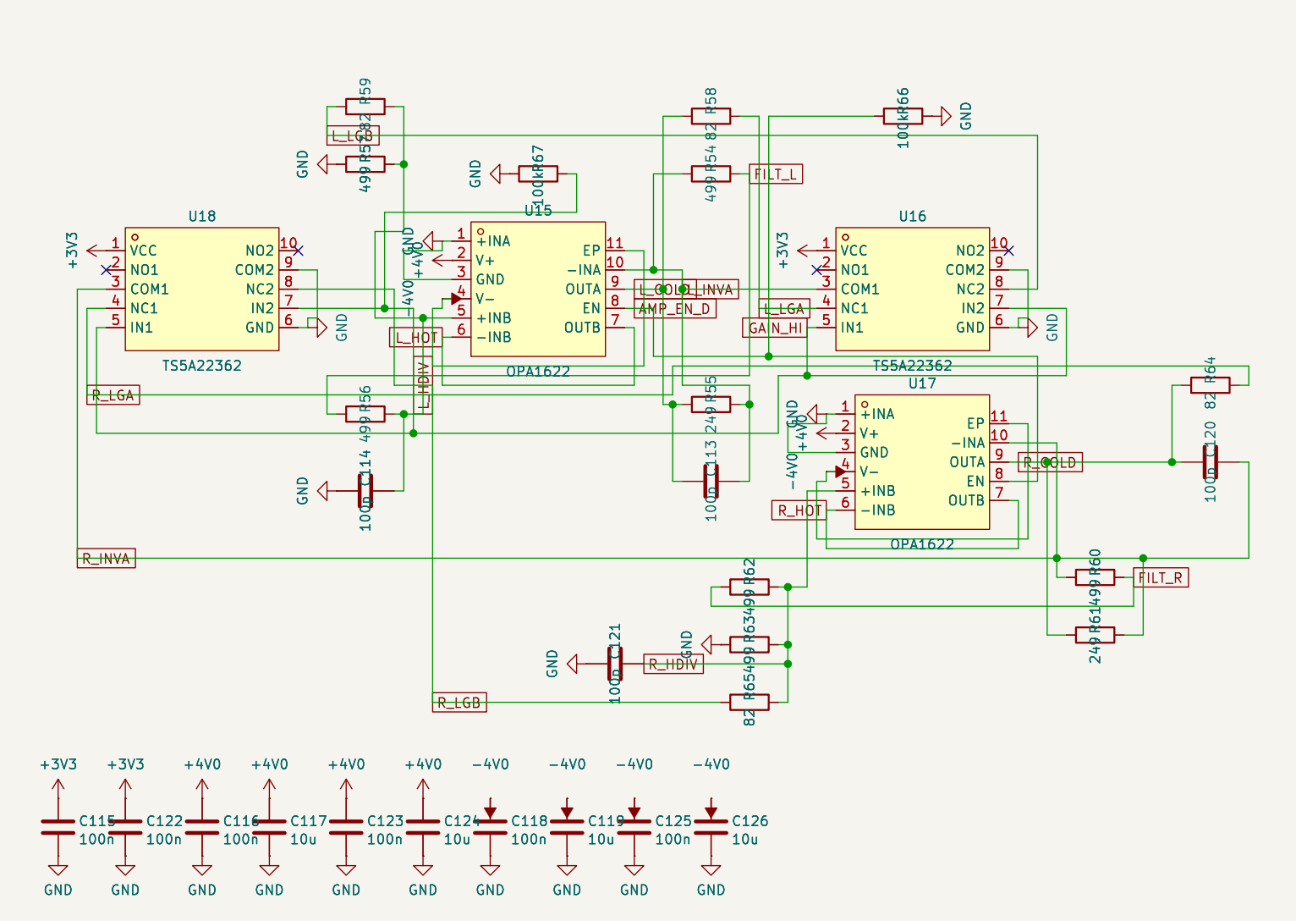
Output filter

AKM Figure 80 per channel: differential to single-ended, 0 dB, fc 133 kHz, Q 0.68, no DC at the output. 2.0 Vrms at 0 dBFS.



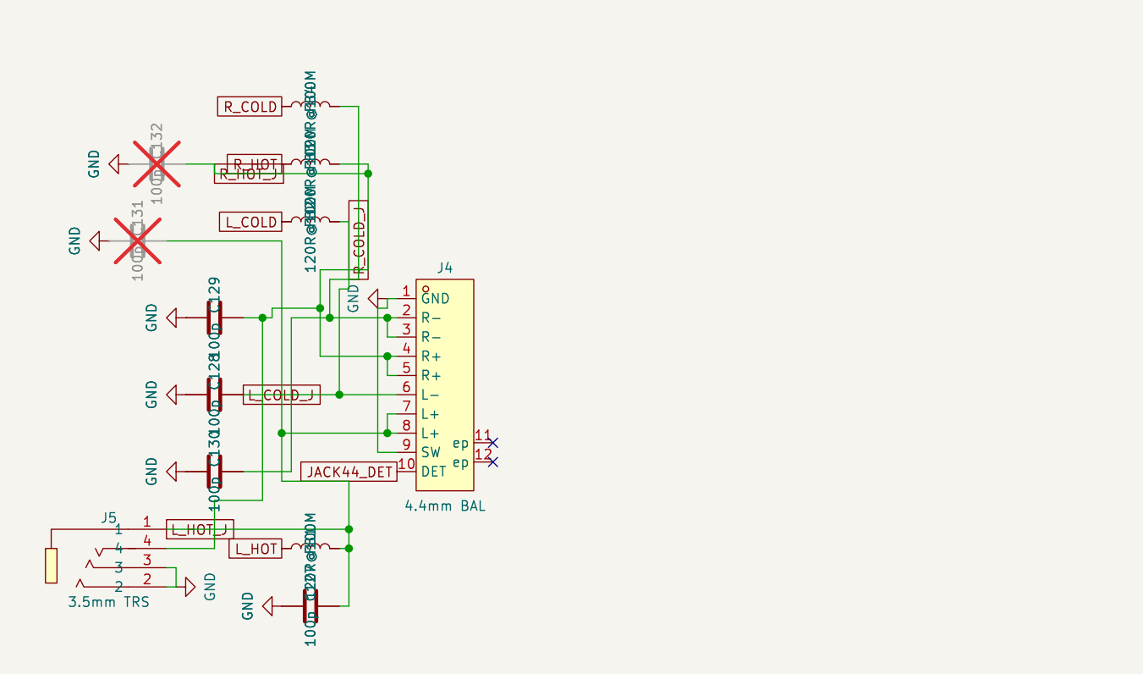
Headphone drivers

OPA1622 per channel. A: -0.5x of the filter output (cold leg). B: follower of the filter output halved by a 499/499 divider (hot leg). 0.94 Vrms per leg, 1.9 Vrms balanced at 0 dBFS. Low gain (-12 dB) when GAIN_HI is low, so the board boots quiet: TS5A22362 switch 1 adds 82R from cold to A's virtual ground, switch 2 adds 82R from the divider tap to ground. Both switch terminals sit at 0 V, so their Ron flatness adds no distortion. EN on its own GPIO: enable after AMP_PG and the DAC's VCM have settled, disable before AMP_EN drops the rails.



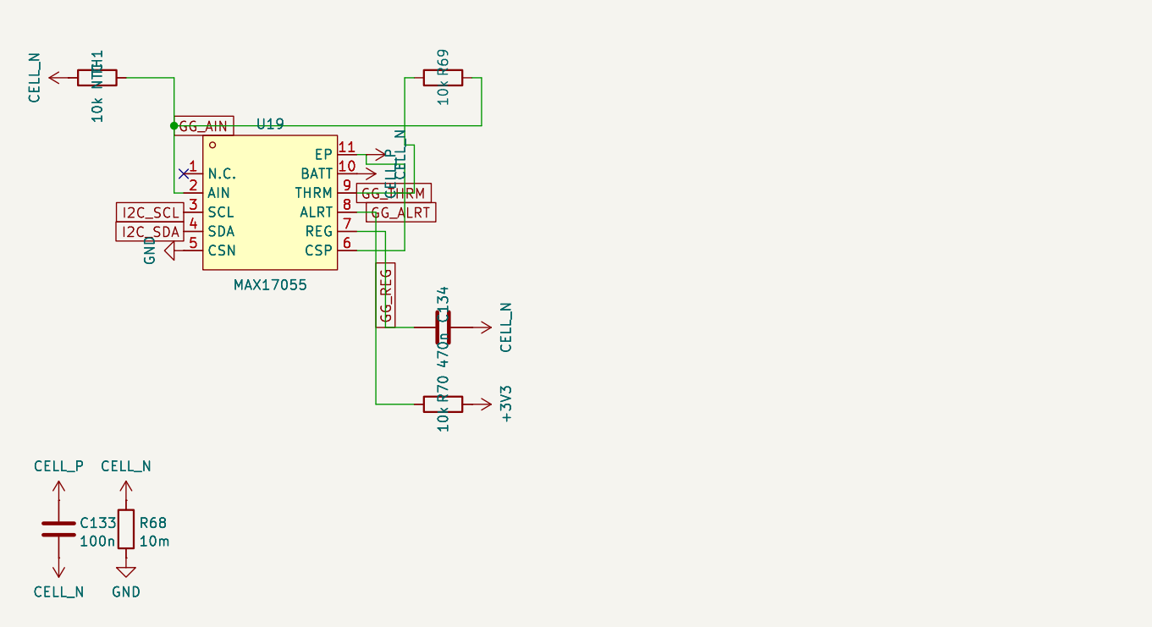
Jacks

4.4 mm: L+ L- R+ R- GND, insert switch to GPIO_03. 3.5 mm TRS from the hot legs (no insert switch on the PJ-320D). Hand-soldered, on the B side.
RF filter: the headphone cable picks up the phone's transmit bursts; beads after the feedback point and 100 pF at each jack pin keep RF out of the driver outputs.
PJ-320D contacts by depth from the mouth (its drawing): D 3.3 mm sleeve, C 6.3 ring 2, B 10.3 ring 1, A 11.4 tip. A TRS plug's ring sits where ring 1 does and its sleeve covers ring 2 and sleeve, so R is on B and both C and D are ground (a TRRS headset's mic grounds too).



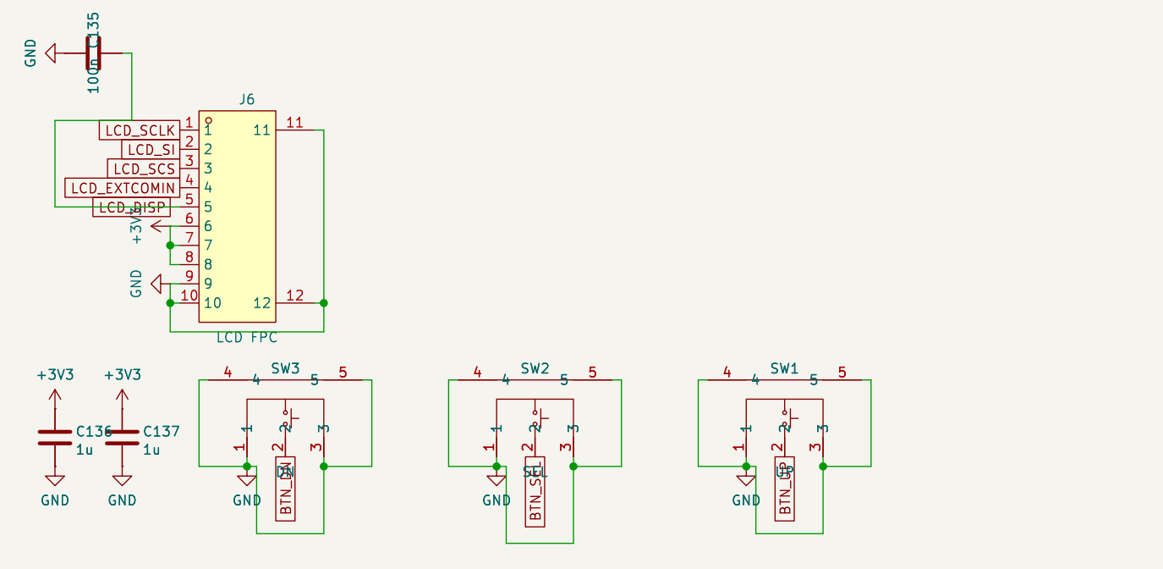
Fuel gauge

MAX17055 (ModelGauge m5 EZ): counts charge through the 10 mOhm in the cell's negative lead and corrects with a cell model, so it stays right while the pod charges the phone. Configured by register writes (no vendor tool), 0x36 on the RT1011's I2C. As Maxim's EV kit: IC ground (CSP, EP) on the cell side of the resistor, CSN on the system side, Kelvin-routed. 10 mOhm: +/-5.12 A range, 160 mW at 4 A. NTC on the board above the cell.



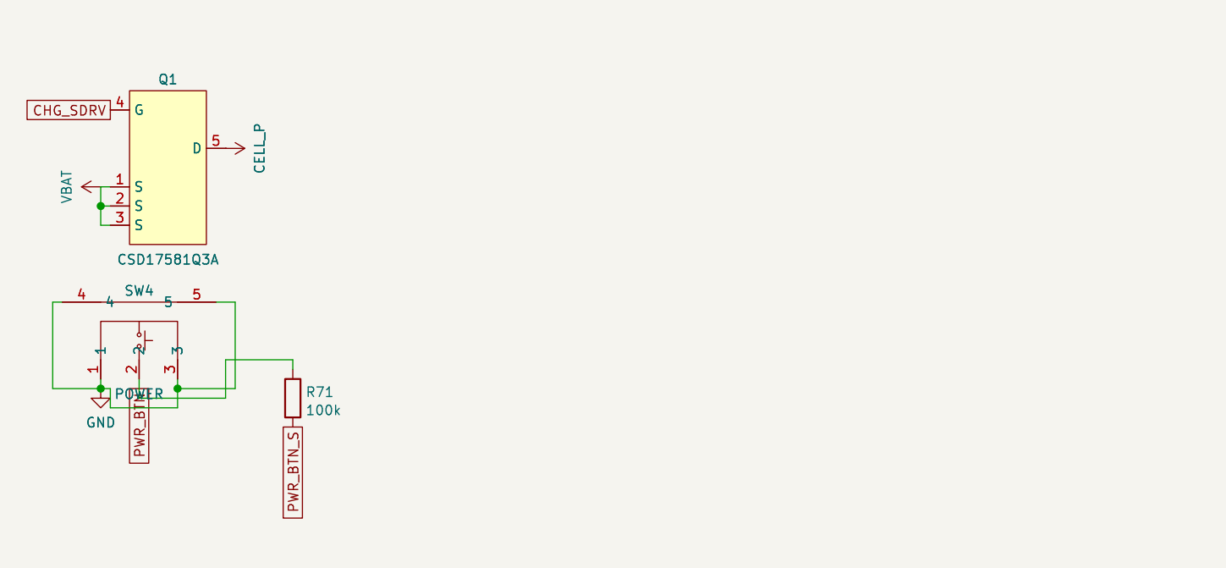
Display and buttons

Sharp LS011B7DH03 memory LCD (1.08 in, 160x68, the nice!view's panel, as on the Corne v3): glued under a window in the pod's outer wall, its FPC folded back under the glass into J (Hirose FH34SRJ, contacts top and bottom, so either FPC face works). EXTMODE high: the RT1011 toggles EXTCOMIN at about 1 Hz, so no SPI traffic while the image holds. Three side buttons on the left edge, internal pull-ups in the RT1011. VDD 3.3 V is the panel's recommended maximum.



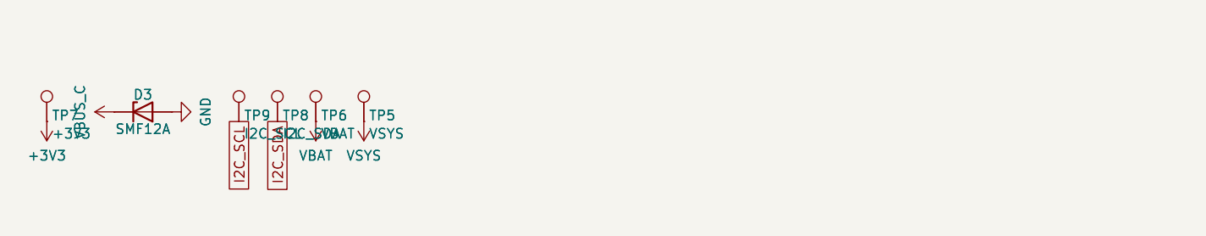
Power button and ship FET

True off: the RT1011 asks the BQ25792 (through the TPS25750's I2Cw) for ship mode, and the ship FET disconnects the cell from everything but the gauge and the charger's sense pin. The power button is on QON (200k pull-up inside the BQ25792): 1 s wakes from ship mode, 10 s power-cycles the whole system (ship FET off 350 ms) whatever the firmware is doing, with or without a charger. The RT1011 reads the button through 100k (GPIO_SD_13) for its own long press. The FET's body diode points into the cell, so with the FET off the cell can't discharge. CSD17581Q3A: about 5 mOhm at the 5 V SDRV drive, 0.1 W at 4.5 A (the pod charging the phone at 3 A).



Bring-up pads and VBUS surge clamp

Added after the physical audit (docs/physical-audit.md). Labelled 1 mm pads for the rails and the I2C bus a first board is probed on, on the outer side, reachable with the lid off.
TVS on VBUS at the connector: SMF12A (12 V stand-off, 14.7 V breakdown, 19.9 V clamp at 10 A), for a charger that attaches above 5 V or rings on a long cable. The pod sinks 9 V at most; a 15 V or 20 V profile needs a higher-voltage TVS first.



Power flags

(c) 2026 Priyanku Baruah. All rights reserved.
Generated by gen_schematic.py from circuit.py: edit those, not this file

Sheet: /
File: proto.kicad_sch

Title: Open Music Player phase 1 prototype

Size: A0
Date:
Rev: A
KiCad E.D.A. 9.0.9
Id: 1/1